Chess Robot Arm: Electronics Design

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This memorandum accompanies the electronics design submission for the Chess Robot Arm. Included with this submission are the following items:

1. Fusion 360 electronics design export (.f3z)
2. Detailed bill of materials (BOM) including:
    - Reference Designators
    - Quantity
    - Manufacturer/vendor part numbers
    - Supplier links
3. Mechanical assembly drawing showing PCB integration into the overall design
4. Updated high-level wiring diagram

The Chess Robot Arm consists of a Xbox-type controller which contains two modes, selected by moving a switch on the PCB; One to manually move the arm with joysticks, and the other to have the robot autonomously play chess with the user using a Raspberry Pi camera system. The first mode is the main goal of the project, while the second mode is the reach goal.

Please refer to the attached documentation for detailed design information, component sourcing, and wiring specifications.